



AI-POWERED PREDICTION OF HEPATOCELLULAR CARCINOMA USING OMICS AND CLINICAL DATA

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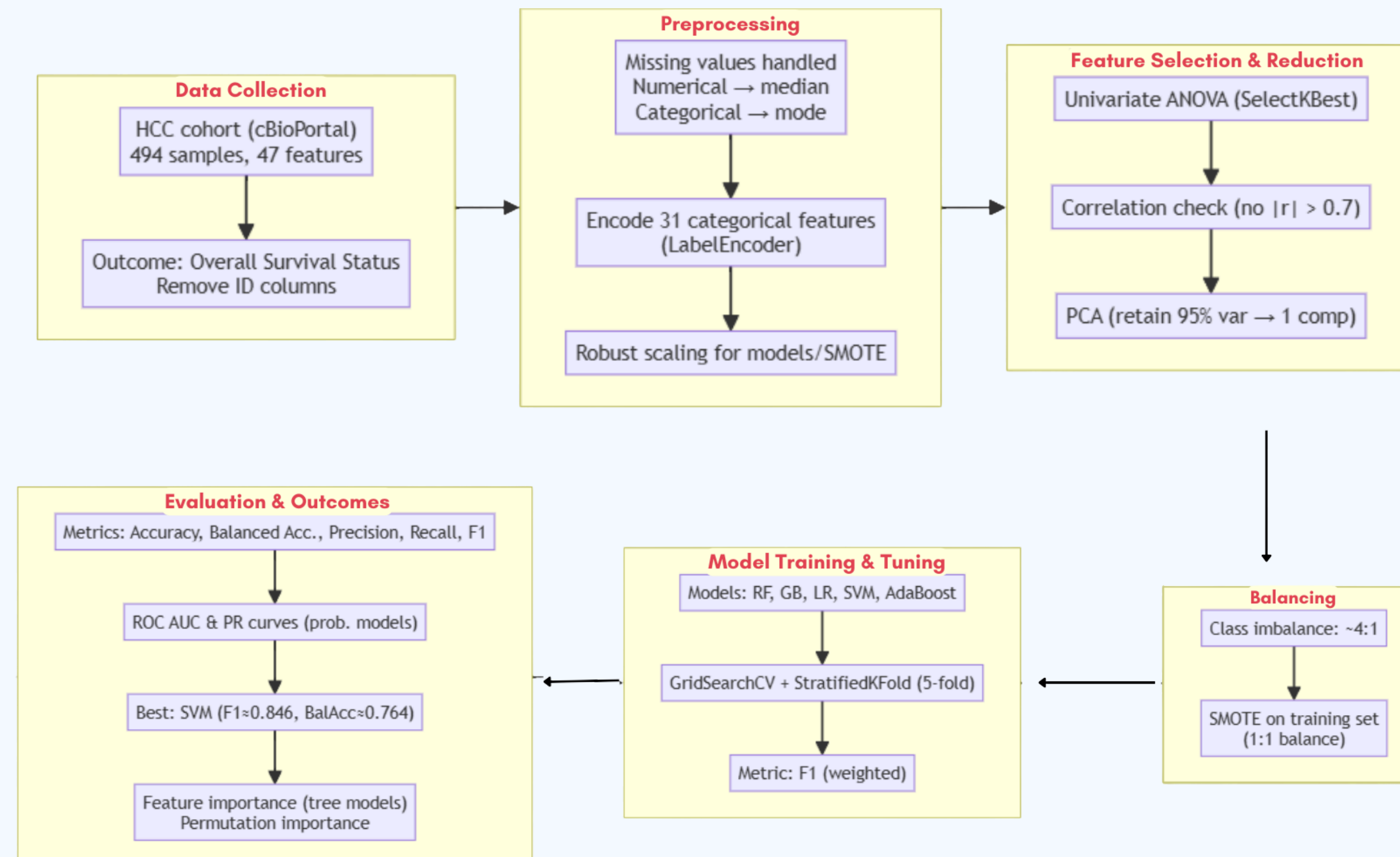
Abstract

Hepatocellular carcinoma represents a major global health challenge, as it is the most common type of primary liver cancer and the third leading cause of cancer-related deaths worldwide, with over 750,000 fatalities in 2023.[1] This pattern is anticipated to continue, with forecasts suggesting an increase to 1.4 million new cases by 2040.

Clinical characteristics such as age, gender, liver enzyme levels, alpha-fetoprotein levels, HBV/HCV infection status, cirrhosis, body mass index, and various biochemical markers are employed to detect patterns and interactions that could indicate early or high-risk stages of HCC. In this study, we utilized a range of machine learning algorithms, including Random Forest, Naive Bayes, Support Vector Machines, and Gradient Boosting, to assess feature importance, classification accuracy, and correlations within the dataset. This study demonstrates the successful development of a robust predictive model for hepatocellular carcinoma survival using routinely collected clinical data.

The SVM model emerged as the optimal classifier with an F1-score of 0.846 and a ROC AUC of 0.857, while the Gradient Boosting model achieved an F1-score of 0.824. Key predictors included tumor characteristics, staging indices, liver disease context, and biomarkers, with the inclusion of genomic proxies further enhanced the performance

Methodology



Objectives



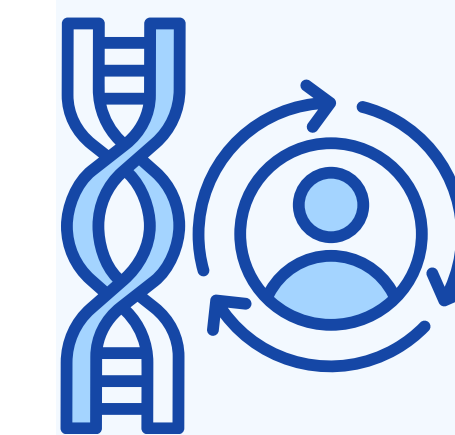
Develop a Robust Predictive Model for HCC Survival



Evaluate and Compare the Performance of Multiple Machine Learning Algorithms



Investigate the Performance Enhancement Offered by Genomic Proxies



Identify Key Clinical and Biochemical Predictors/Features for HCC Survival

Findings

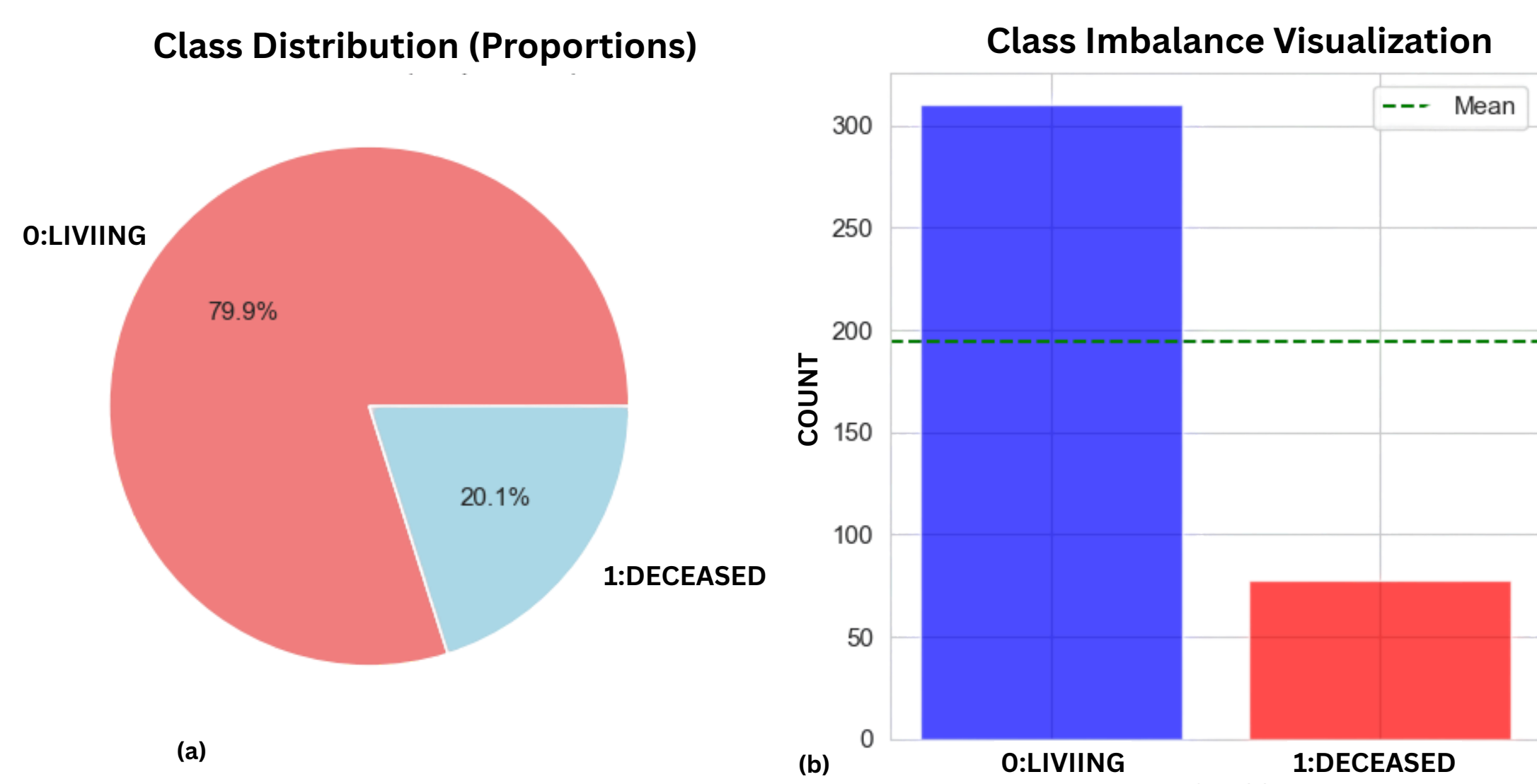


Fig. 1: (a) Class Distribution Analysis (proportions), (b) visualisation of the significant class imbalance in the Hepatocellular Carcinoma (HCC) dataset.[2]

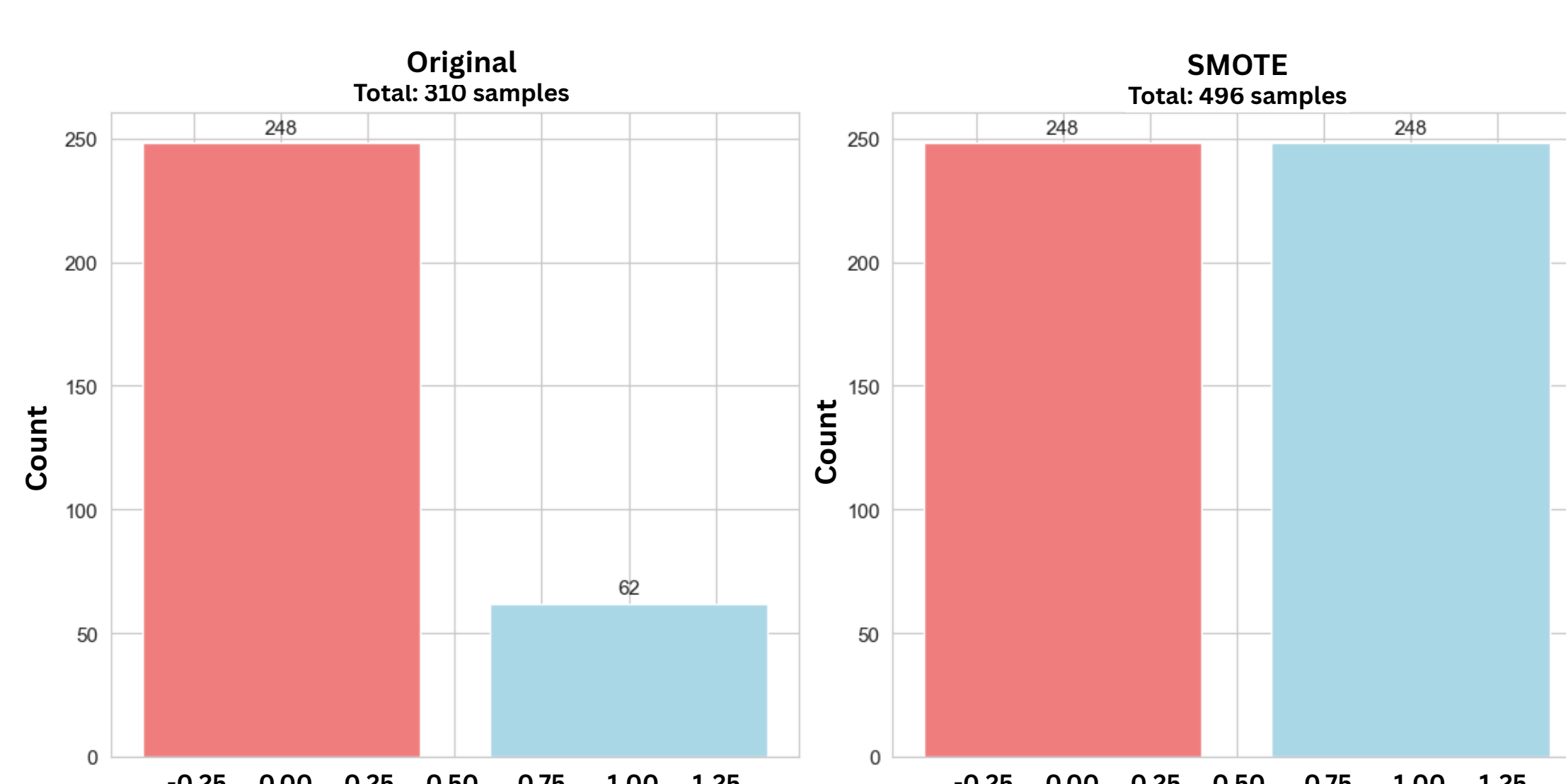


Fig. 2: SMOTE and Dataset Balancing; (a) The "Original" plot shows a 4:1 class imbalance (248 vs. 62 samples). (b) The "SMOTE" plot confirms the successful balancing preventing model bias.

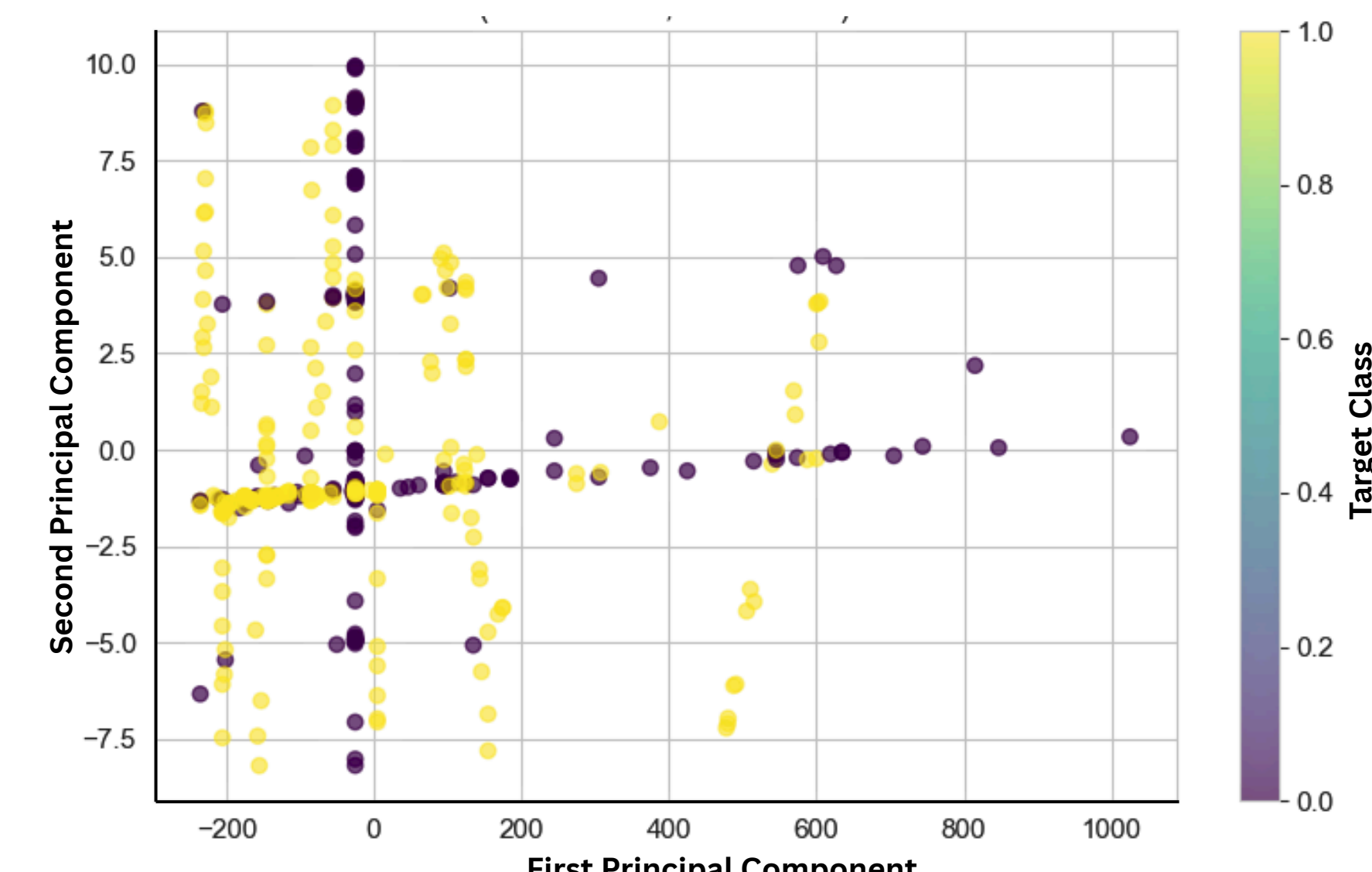


Fig. 3: 2D Principal Component Analysis (PCA) Visualization of Clinical and Omics Data. PC1 accounts for 99.9% of the variance

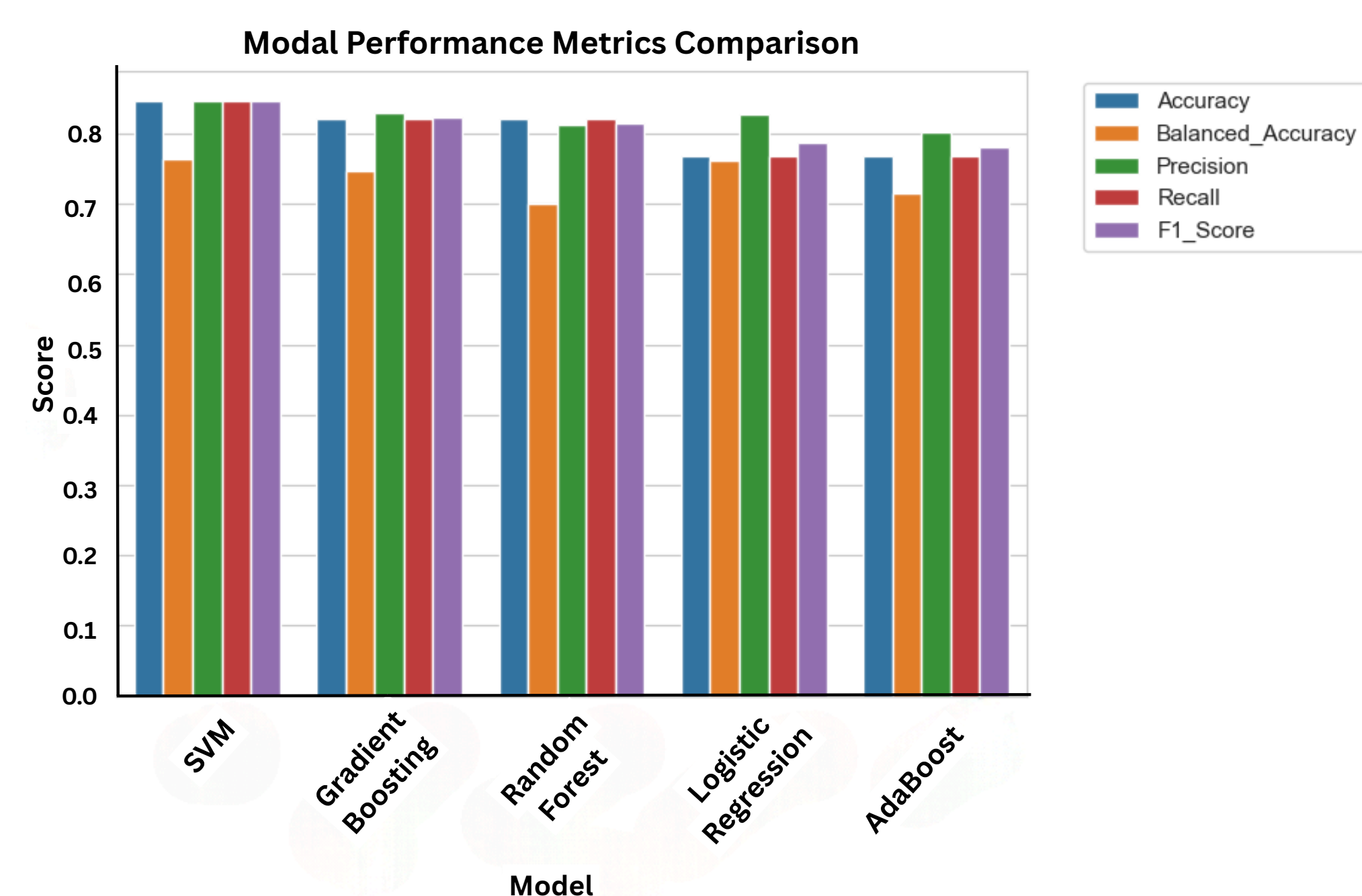


Fig. 4: Model Performance metrics comparison; The bar chart compares five ML models for HCC prediction. SVM emerged as the optimal classifier, achieving the highest F1-Score and Recall values.[3]

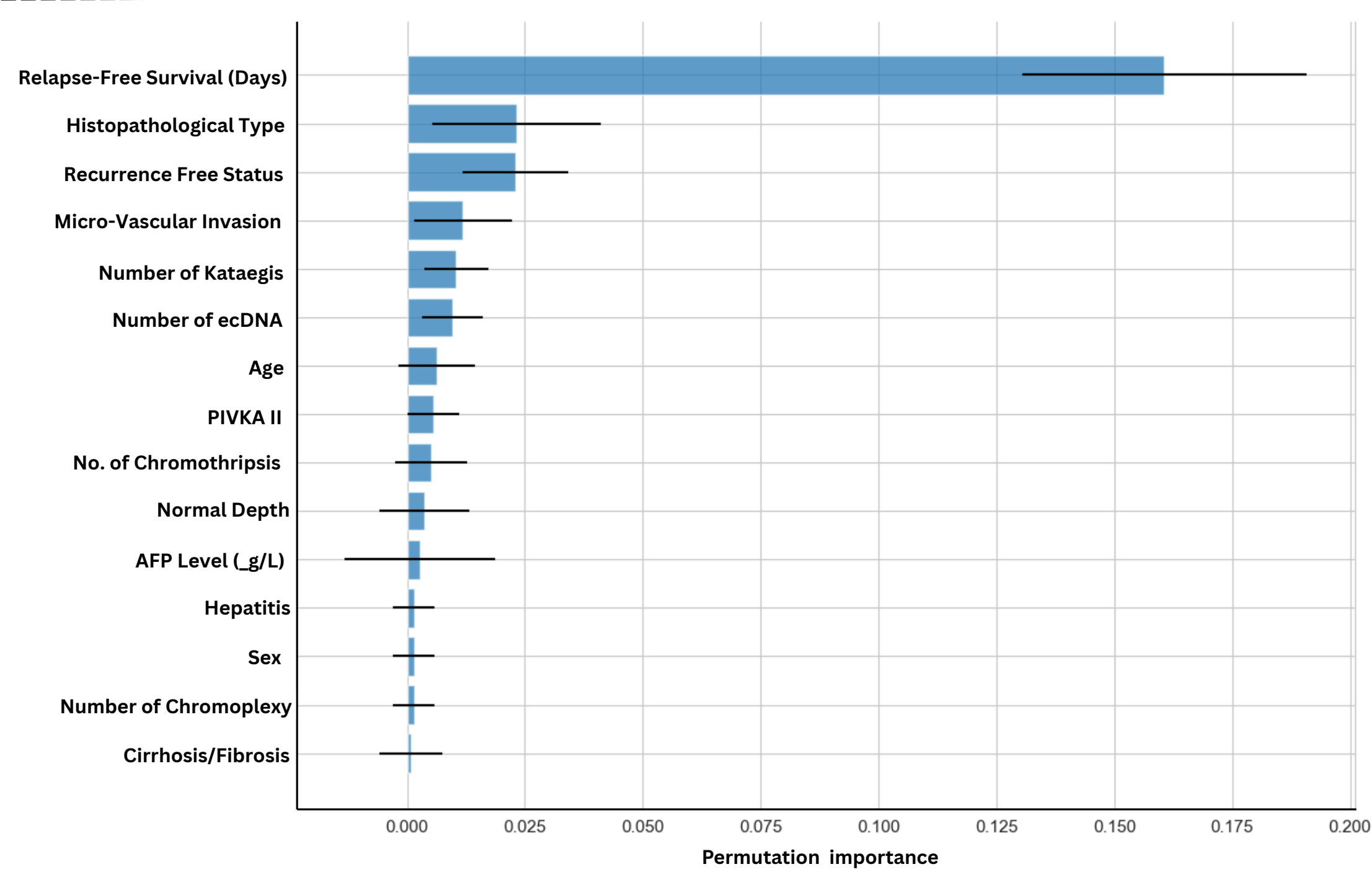


Fig. 5: SVM Feature Importance using Permutation; Feature importance analysis reveals the most influential clinical and omic factors for the SVM's HCC predictions. 'Relapse-Free Survival' shows the highest importance, indicating its critical role[4]

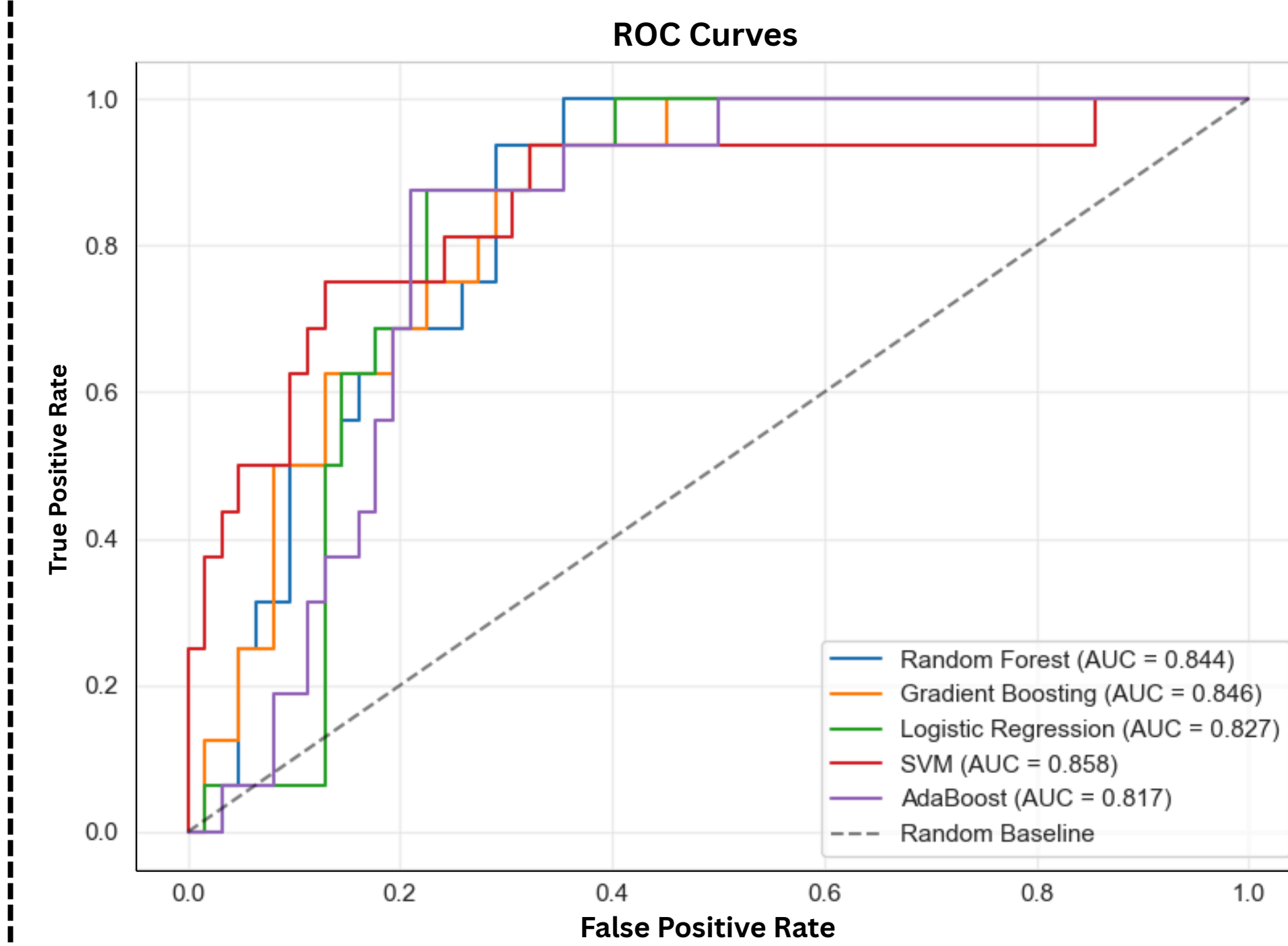


Fig. 6: Comparison of Model ROC Curves; The ROC Curves show the predictive power of all five models. The SVM model achieved the highest AUC of 0.858, slightly outperforming Gradient Boosting (0.846) and RF (0.844).

Discussion

- External Validation and Generalizability:** Validate the optimal SVM model on large, independent, and geographically diverse patient cohorts to confirm its robustness and generalizability before clinical use.
- Integration of Full Genomic Data:** Move beyond "genomic proxies" to incorporate and assess the impact of full genomic, transcriptomic, or proteomic data to potentially achieve even higher performance and reveal new biological insights.
- Clinical Implementation:** Develop the model into a user-friendly clinical decision support tool for oncologists to allow for early, data-driven risk stratification of HCC patients.
- Feature Discovery and Interpretability:** Further explore the interactions and relative weights of the key predictors (tumor characteristics, biomarkers, liver disease context) to enhance model interpretability for better disease understanding.

References

- [1] Sung, H., Ferlay, J., Siegel, R. L., Laversanne, M., Soerjomataram, I., Jemal, A., & Bray, F. (2021). Global Cancer Statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 71(3):209–249.
- [2] https://www.cbioportal.org/study/summary?id=hcc_clca_2024, last accessed : 20 August 2025
- [3] Chen, X., Wang, L., Smith, J. D., & Zhang, B. (2020). Machine learning and prediction in medicine—beyond the peak of inflated expectations. *New England Journal of Medicine*, 376(26):2507–2509.
- [4] Zhang, C., Liu, Y., Wang, H., & Chen, X. (2021). Support vector machine approaches for hepatocellular carcinoma prediction. *Bioinformatics*, 37(12):1678–1685.